

4 Algebraic structures

Introductory problem

What information can be extracted from the equation $ax + b = 3x + 2$?

What information can be extracted from the identity $ax + b = 3x + 2$, for all x ?

Equations are the building blocks of mathematics. There are many different types; some have no solutions, some have many solutions. Some have solutions which cannot be expressed in terms of any function you have met.

Graphs are an alternative way of expressing a relationship between two **variables**. If you understand the connection between graphs and equations, and can switch between the two representations, you will have a wider variety of tools to solve mathematical problems. The International Baccalaureate® places great emphasis on using graphical calculators to analyse graphs.

Inequalities share many of the same properties as equations. However, a few important differences mean you need different techniques to solve some inequalities.

In much of mathematics we do not distinguish between equations and **identities**, although they are fundamentally different. In this chapter we shall explore some of these differences and look at the different techniques we can apply.

4A Solving equations by factorising

We start by looking at some common algebraic methods for solving equations. You have already used factorising to sketch polynomials, and in this chapter we will apply the same principles in a wider context.

In this chapter you will learn:

- some standard algebraic strategies for solving equations
- how to sketch graphs, and some limitations of graphical calculators
- how to use a graphical calculator to solve equations
- how to solve systems of up to three linear simultaneous equations
- algebraic and graphical strategies for solving inequalities
- how to prove and work with identities.

This is true for real numbers, but if you know that the numbers you are looking for are integers, then knowing the product of two numbers is far more useful.



If two numbers multiply together to give 5, what can you say about those two numbers?

They could be 1 and 5, 10 and $\frac{1}{2}$, π and $\frac{5}{\pi}$; there are actually an infinite number of possibilities. So if you know that two numbers multiply together to give 5 it does not help to find what those numbers are. However, if two numbers multiply together to give zero, this is only possible if one of them is zero.

If we can factorise an expression which is equal to zero then this is a very powerful tool for solving equations.

Worked example 4.1

Solve the equation $e^x(\ln(x) - 1)(2x - 1) = 0$.

Set the three factors equal to 0 then solve each separately

$$\begin{aligned} \text{Either } e^x &= 0 & (1) \\ \text{or } \ln(x) - 1 &= 0 & (2) \\ \text{or } 2x - 1 &= 0 & (3) \end{aligned}$$

From (1)

$e^x = 0$ has no solution

From (2)

$$\begin{aligned} \ln(x) &= 1 \\ \Leftrightarrow x &= e \end{aligned}$$

From (3)

$$\begin{aligned} 2x &= 1 \\ \Leftrightarrow x &= \frac{1}{2} \end{aligned}$$

List the solutions

$$x = e \text{ or } \frac{1}{2}$$

EXAM HINT

If you divide by a function instead of looking at factors then it is possible that you might lose solutions. For example, consider the equation $x^3 = x$. If you divide both sides by x you get $x^2 = 1$ so $x = \pm 1$. However, if you put $x = 0$ into the original equation you will see that this is also a solution. The correct method for solving this equation is by factorisation.

$$\begin{aligned} x^3 &= x \\ \Rightarrow x^3 - x &= 0 \\ \Rightarrow x(x^2 - 1) &= 0 \\ \Rightarrow x(x - 1)(x + 1) &= 0 \\ \Rightarrow x = 0 \text{ or } x = 1 \text{ or } x = -1 \end{aligned}$$

Exercise 4A

1. Solve the following equations:
- (a) (i) $3(x-3)^3 = 0$ (ii) $-4(x+1)^5 = 0$
 (b) (i) $7(2x-1)(5x+3)^2 = 0$ (ii) $5(3-x)^2(2x+6)^2 = 0$
 (c) (i) $(\log_3 x - 3)(3^x - 3) = 0$ (ii) $(\sqrt{x} - 4)(9\sqrt{x} - 1) = 0$
 (d) (i) $x(x^2 - 3) = 7(x^2 - 3)$ (ii) $5x(x^2 - 5x + 4) = 6(x^2 - 5x + 4)$
 (e) (i) $6^x - 4 \times 3^x = 0$ (ii) $2 \times 5^x - 7 \times 10^x = 0$

2. Solve the following equations:

- (a) (i) $x^3 + 12 = 2x^2 + 11x$
 (ii) $x^3 - x^2 - 17x = 15$
 (b) (i) $x^3 - 5x^2 + 7x - 2 = 0$
 (ii) $x^3 - 6x^2 + 7x - 2 = 0$

3. (a) Find the roots of these equations:

(i) $x^3 - 6x^2 + 11x = 6$ (ii) $x^3 - 2x^2 + 6 = 5x$

- (b) Find the zeros of these polynomials:

(i) $x^3 + x^2 - x - 1$ (ii) $x^3 - 3x^2 - 10x + 24$

4. Solve $(3x-1)^{x^2-4} = 1$.

[4 marks]

5. Solve the equation $x|x| = 4x$.

You saw how to use the factor theorem to factorise polynomials in Section 3B.

4B Solving equations by substitution

There are certain types of equation which you should know how to solve; in this section we shall focus particularly on quadratic equations as we have a formula to solve them.

We shall also see how a substitution can turn some complicated-looking equations into quadratic equations.

Worked example 4.2

Solve the equation $x^4 - 4 = 3x^2$.

A substitution $y = x^2$ turns this into a quadratic equation since $x^4 = y^2$

If $y = x^2$ the equation becomes $y^2 - 4 = 3y$

continued...

This is a standard quadratic equation

Use the substitution to find x

$$y^2 - 3y - 4 = 0$$

$$\Rightarrow (y+1)(y-4) = 0$$

$$\Rightarrow y = -1 \text{ or } y = 4$$

$$\text{Therefore } x^2 = 4 \text{ or } x^2 = -1 \text{ (reject)}$$

$$\Rightarrow x^2 = 4$$

$$x = 2 \text{ or } x = -2$$

Other substitutions may not be so obvious.

In particular, it is quite common to be given an exponential equation which needs a substitution. Look out for an a^x and an a^{2x} or an $(a^2)^x$ term, both of which can be rewritten as $(a^x)^2$.

Worked example 4.3

Solve the equation $2(4^x + 2) = 9 \times 2^x$.

A substitution $y = 2^x$ turns this into a quadratic equation since $4^x = 2^{2x} = y^2$

This is a standard quadratic equation

Use the substitution to find x

If $y = 2^x$ the equation becomes

$$2(y^2 + 2) = 9y$$

$$2y^2 - 9y + 4 = 0$$

$$\Rightarrow (2y-1)(y-4) = 0$$

$$\Rightarrow y = \frac{1}{2} \text{ or } y = 4$$

$$\text{Therefore } 2^x = \frac{1}{2} \text{ or } 2^x = 4$$

$$\Rightarrow x = -1 \text{ or } x = 2$$

Exercise 4B

1. Solve the following equations, giving your answers in an exact form.

(a) (i) $a^4 - 10a^2 + 21 = 0$

(ii) $x^4 - 7x^2 + 12 = 0$

(b) (i) $2x^6 + 7x^3 = 15$

(ii) $a^6 + 7a^3 = 8$

(c) (i) $x^2 - 4 = \frac{2}{x^2}$

(ii) $x^2 + \frac{36}{x^2} = 12$

- (d) (i) $x - 6\sqrt{x} + 8 = 0$ (ii) $x - 10\sqrt{x} + 24 = 0$
 (e) (i) $e^{2x} + 16e^x = 80$ (ii) $e^{2x} - 9e^x + 20 = 0$
 (f) (i) $25^x - 15 \times 5^x + 50 = 0$ (ii) $4^x - 7 \times 2^x + 12 = 0$
 (g) (i) $\log_4 x = (\log_4 x^2)^2$ (ii) $(\log_3 x)^2 - 3\log_3 x + 2 = 0$

2. Solve the equation $9(1 + 9^{x-1}) = 10 \times 3^x$. [5 marks]

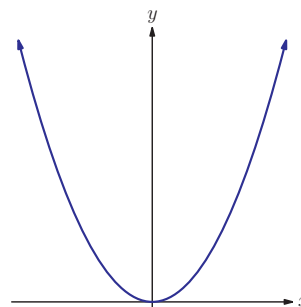
3. If $a^x = 6 - \frac{5}{a^x}$, where $a > 0$, solve for x , leaving your answer in terms of a . [5 marks]

4. Solve the equation $\log_2 x = 6 - 5\log_x 2$. [6 marks]

4C Features of graphs

You will meet many different types of equations in this course, and will learn various techniques for solving them. In Chapters 2 and 3 you have already met polynomial, exponential and logarithmic equations. However, you also have to be able to find approximate solutions to equations which are not from among the standard types. This can be done using graphs.

Graphs are simply another way of representing a relationship between two variables. For example, we can write $y = x^2$ or draw the graph alongside.



In this section we shall examine important features you should be looking for when sketching graphs using your graphical calculator, and also discuss some limitations of this method.

The main features you should indicate on your sketch are:

- y-intercept; this is where $x = 0$
- x-intercepts (zeros); these are where $y = 0$
- maximum and minimum points.

See Calculator skills sheets 2 and 4 on the CD-ROM about sketching graphs and finding the main features.

If the graph you are sketching is completely unfamiliar, it can be difficult to choose a good viewing window. If the window is too small you may be missing important parts of the graph, and if it is too large you may not be able to distinguish between features that are close together. Having an idea about the overall shape of the graph is very useful, so it is important to learn about different types of graphs, even if you have a graphical calculator available.

The x- and y-intercepts were discussed in more detail in chapter 3.



Worked example 4.4

Sketch the graph of $y = \frac{x^3 - 16x}{x^2 + 1}$.

Find the important features

We know that the zeros will occur when the numerator is zero, and we can find those points by factorising

The viewing window should include 4 and -4

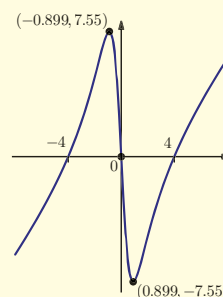
There appear to be one maximum and one minimum point; we should find their coordinates

$$x^3 - 16x = 0$$

$$\Leftrightarrow x(x^2 - 16) = 0$$

$$\Leftrightarrow x(x - 4)(x + 4) = 0$$

$$\Leftrightarrow x = 0, 4, -4$$



Some graphs might have asymptotes.

KEY POINT 4.1

An **asymptote** is a straight line which the graph approaches as either x or y gets very large.

Asymptotes are usually shown on graphs by dotted lines.

Vertical asymptotes occur where a function ceases to be defined.

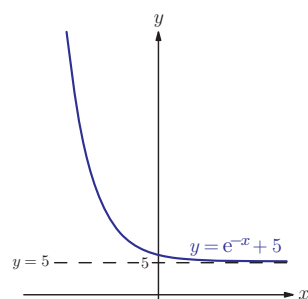
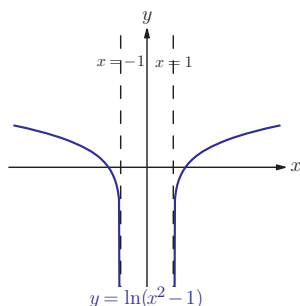
They are vertical lines of the form $x = a$.

For example, $y = \frac{1}{x - 3}$ has an asymptote $x = 3$ (because we

cannot divide by zero), and $y = \ln(x^2 - 1)$ has asymptotes $x = -1$ and $x = 1$ (because we cannot take a logarithm of zero).

Horizontal asymptotes indicate long term behaviour of the graph and are lines that the graph approaches for large values of x (positive or negative). They are horizontal lines of the form $y = a$. For example, $y = e^{-x} + 5$ has an asymptote $y = 5$.

Since the asymptotes are not a part of the graph, they will not show on your calculator sketch. Using the cursor allows you to investigate what happens for large values of x or y , so you can find the approximate positions of the asymptotes. To find their exact equations you need to apply some knowledge of the



function in question; for example, knowing that you cannot divide by zero, or that e^{-x} approaches zero for large x . You will only be asked to identify the exact position of asymptotes for familiar functions.

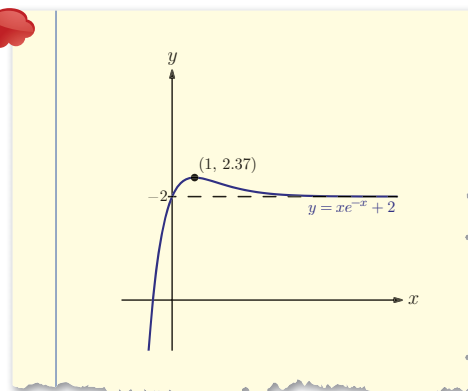
Worked example 4.5

Sketch the graph of $y = xe^{-x} + 2$.

Start with a standard window with both x and y between -10 and 10

It looks like there is something interesting happening for x values around 1 , so zoom in a bit to confirm that there is a maximum point

There also seems to be an asymptote; we know that e^{-x} approaches 0 , so it is likely that the asymptote is $y = 2$



Note that although the line $y = 2$ is an asymptote, the graph actually crosses it when $x = 0$. This is fine, as the asymptote is only relevant for large values of x .

Vertical asymptotes can sometimes be unclear. In the following example the graph approaches the vertical asymptote from two sides, so the calculator may attempt to 'connect the points' and form a 'V' shape. You must identify for yourself that this is an asymptote and draw it correctly.

See Prior learning Section I on the CD-ROM for an introduction to the modulus function $|x|$. Make sure you know where to find it on your calculator.

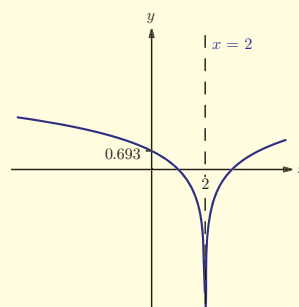


Worked example 4.6

Sketch the graph of $y = \ln |x - 2|$.

It looks like the two branches of the graph join at a point. However, we know that $\ln 0$ is not defined, so there should be an asymptote at $x = 2$

Asymptote: $x - 2 = 0 \Leftrightarrow x = 2$



Exercise 4C



1. Sketch the following graphs, indicating all the important features: axis intercepts, vertical and horizontal asymptotes, maximum and minimum points.

(a) (i) $y = x^4 - x^3 + 1$ (ii) $y = x^4 - x^2$

(b) (i) $y = (x-1)e^x$ (ii) $y = (e^x - 1)^2$

(c) (i) $y = \frac{\frac{1}{2}e^x - 1}{x-1}$ (ii) $y = \ln\left(\frac{x+2}{x-1}\right)^2$

(d) (i) $\left|\frac{x^2-1}{x+2}\right|$ (ii) $y = \frac{|x^2-4|}{x+1}$

For questions 2 to 4, mark the coordinates of all zeros, maximum and minimum points.



2. Sketch the graph of $y = x \ln x$. [4 marks]



3. Sketch the graph of $y = \frac{e^x}{\ln x}$. [6 marks]



4. Sketch the graph of $y = \frac{x^2(x^2-9)}{e^x}$ for $-5 \leq x < 10$. [6 marks]

4D Using a graphical calculator to solve equations

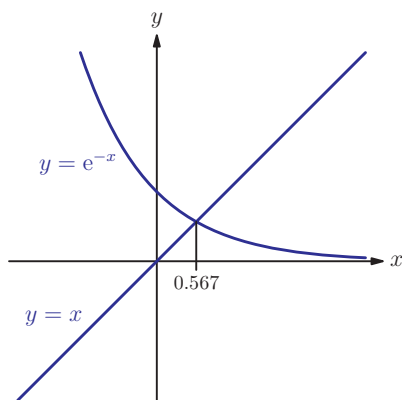
Some equations cannot be solved analytically. This means that you cannot rearrange them to get $x = \text{value}$ using the standard set of functions mathematicians allow. However, there are still values of x which satisfy the equation.

An example of such an equation is $x = e^{-x}$.

One good way to find these solutions is by plotting both sides of the equation using a graphical calculator. Calculator skills sheet 5 explains how to find the intersection of two graphs.

The solution to the equation $x = e^{-x}$ can actually be written in terms of the 'Lambert W Function'. In fact, it is $W(1)$. However, does knowing this actually give us any more information about the solution to the equation?





EXAM HINT

When using your calculator to solve an equation, you must draw a sketch of the graph and round the solution to an appropriate degree of accuracy (usually 3 significant figures).

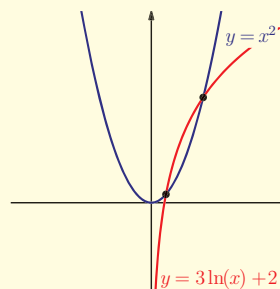
The x -value of the intersection gives the solution to the equation, in this case 0.567 to 3 significant figures.

Worked example 4.7

Solve the equation $x^2 = 3\ln x + 2$.

There is no obvious substitution or factorisation, so plot on the calculator and sketch

Use the calculator to find the intersection points



$x = 2.03$ or 0.573 (3SF) from GDC

Solving equations graphically has some problems: You may not know how many solutions you are looking for, or how to set the viewing window so you can see them all. If two solutions are very close together you may miss one of them. This is where you need your knowledge of the shapes of different graphs to make sure that the calculator is showing all the important features.

Many graphical calculators have functions for solving some special types of equations. In particular, you may be able to solve polynomial equations (those involving only positive integer powers of x) without having to use a graph.

EXAM HINT

If the question does not ask for an exact answer this is an indication that a graphical solution might be appropriate. If you cannot see another quick way to do it, put it into your calculator.

You may also be able to use an equation solver tool, although this has similar issues to using graphs. See Calculator skills sheets 5 and 6 on the CD-ROM for more details.



Worked example 4.8

Solve the equation $x^3 = 5x^2 - 2$.

We don't know any algebraic methods for solving cubic equations, so use the polynomial equation solver on the calculator

The equation needs to be rearranged

$$\begin{aligned} x^3 - 5x^2 + 2 &= 0 \\ \Rightarrow x &= -0.598, 0.680, 4.92 \text{ (3SF)} \\ &\text{(using GDC)} \end{aligned}$$

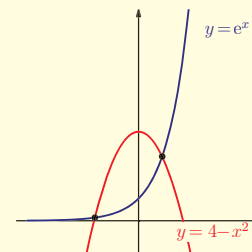
EXAM HINT

You can use the graphical method in the non-calculator paper, too. In such questions you are often just asked to find the number of solutions, rather than actually solve the equation.

Worked example 4.9

Find the number of solutions of the equation $e^x = 4 - x^2$.

We know how to sketch both graphs



The solutions correspond to the intersections of the two curves

There are 2 solutions.

Exercise 4D

1. Solve the following equations giving your answers to 3 significant figures.

(a) (i) $x^3 = 3x - 1$

(ii) $x^3 + 4x^2 = 2x - 1$

(b) (i) $e^x = x + 1$

(ii) $e^{2x} = x^2 - 3$

(c) (i) $e^x = \ln x$

(ii) $e^x \ln x = x^3 - x$

2. Solve the equation $x \ln x = 3 - x^2$.

[4 marks]

3. The equation $\ln x = kx$ where $k > 0$ has one solution.

How many solutions do the following equations have?

(a) $\ln x^2 = kx$ (b) $\ln\left(\frac{1}{x}\right) = kx$ (c) $\ln\sqrt{x} = kx$ [6 marks]

4E Solving simultaneous equations by substitution

In the previous section we saw how to solve equations by finding intersections of graphs. For example, to solve the equation $e^x = 4 - x^2$ we sketched the graphs of $y = e^x$ and $y = 4 - x^2$ and found their intersections. We were only interested in the x -coordinates of the intersection points. However, the y -values for the two equations are also equal at those points. This means that we have in fact solved simultaneous equations

$$\begin{cases} y = e^x \\ y = 4 - x^2 \end{cases}$$

A powerful method for solving simultaneous equations is **substitution**, where you replace all the occurrences of one of the variables in one equation by an expression from the other equation.

See Prior learning section Q for a reminder about linear simultaneous equations.

Worked example 4.10

Find the coordinates of the points of intersection between the line $2x - y = 1$ and the parabola $y = x^2 - 3x + 5$.

At the intersection points, the y -coordinates for the two curves are equal, so we can replace y in the first equation by the expression for y from the second equation

This rearranges to a quadratic equation
Try to factorise

We also need to find the y -coordinates, by substituting back into one of the equations for y (they will both give the same answer). Pick the first equation, as it is easier

$$2x - y = 1 \Rightarrow y = 2x - 1$$

$$x^2 - 3x + 5 = 2x - 1$$

$$x^2 - 5x + 6 = 0$$

$$(x - 2)(x - 3) = 0$$

$$x = 2 \text{ or } 3$$

$$y = 2x - 1$$

$$x = 2: y = 2 \times 2 - 1 = 3$$

$$x = 3: y = 2 \times 3 - 1 = 5$$

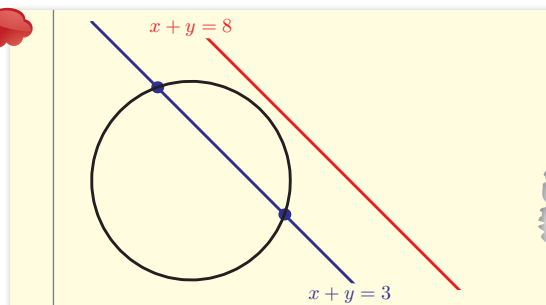
The coordinates are $(2, 3)$ and $(3, 5)$

Sometimes we only want to know how many intersection points there are, rather than to find their actual coordinates. The discriminant can be used to determine the number of intersections if the equation is quadratic.

Worked example 4.11

Find the set of values of k for which the line with equation $x + y = k$ intersects the circle with equation $x^2 - 4x + y^2 + 6y = 12$ at two distinct points.

The question suggests that if we try to find the intersections between the line and the circle, the number of solutions will depend on the value of k . This makes sense, as changing k moves the line up and down, so it will sometimes intersect the circle and sometimes won't



continued...

Try finding the intersections in terms of k and see if that gives any ideas

At the intersection points, the y -coordinates for the two curves are equal, so we can replace y in the second equation by the expression for y from the first equation

This is a quadratic equation; write it with one side equal to zero

We know that the discriminant tells us the number of solutions of a quadratic equation

This is a quadratic inequality. To solve it, find where $\text{LHS} = 0$ and sketch the graph

The graph shows that the required interval is between the roots

Line equation: $y = k - x$

Substitute into the circle equation:

$$x^2 - 4x + (k - x)^2 + 6(k - x) = 12$$

$$\Rightarrow x^2 - 4x + k^2 - 2kx + x^2 + 6k - 6x = 12$$

$$\Rightarrow 2x^2 - (10 + 2k)x + k^2 + 6k = 12$$

$$\Rightarrow 2x^2 - (10 + 2k)x + (k^2 + 6k - 12) = 0$$

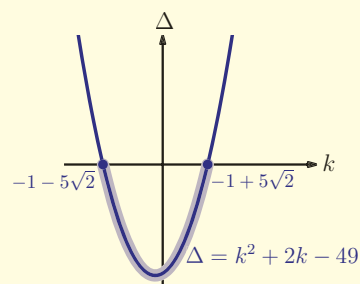
Two solutions $\therefore \Delta > 0$

$$\Delta = (10 + 2k)^2 - 8(k^2 + 6k - 12) > 0$$

$$\Rightarrow 100 + 40k + 4k^2 - 8k^2 - 48k + 96 > 0$$

$$\Rightarrow -4k^2 - 8k + 196 > 0$$

$$\Rightarrow k^2 + 2k - 49 < 0$$



Roots: $k^2 + 2k - 49 = 0$

$$k = \frac{-2 \pm \sqrt{4 + 4 \times 49}}{2}$$

$$= \frac{-2 \pm 2\sqrt{1 + 49}}{2}$$

$$= -1 \pm \sqrt{50} = -1 \pm 5\sqrt{2}$$

$$\therefore -1 - 5\sqrt{2} < k < -1 + 5\sqrt{2}$$

EXAM HINT

You are not expected to know why the second equation represents a circle, or what the centre and the radius of the circle are. Exam questions sometimes involve unfamiliar concepts, but you will be given all the information required to answer the question.

The last example illustrates how a geometrical problem can be solved using purely algebraic methods.



There is a whole branch of mathematics studying such methods, called analytic geometry. It was developed in the 17th Century by the French philosopher and mathematician René Descartes. Establishing a link between geometry and algebra was a major step in the development of modern mathematics. You will learn more about it in chapter 14.

The circle and the parabola belong to a family of quadratic curves called conic sections, which also includes the ellipse and the hyperbola. There are many fascinating and beautiful results concerning conic sections, many of which can be investigated using the properties of the quadratic function.



Exercise 4E



1. Find the coordinates of intersection between the given parabola and the given straight line.

- (a) (i) $y = x^2 + 2x - 3$ and $y = x - 1$
 (ii) $y = x^2 - 4x + 3$ and $y = 2x - 6$
 (b) (i) $y = -x^2 + 3x + 9$ and $2x - y = 3$
 (ii) $y = x^2 - 2x + 8$ and $x - y = 6$



2. Solve these simultaneous equations.

- (a) (i) $x - 2y = 1$ (ii) $x + 2y = 3$
 $3xy - y^2 = 8$ $y^2 + 2xy + 9 = 0$
 (b) (i) $xy = 3$ (ii) $x + y + 8 = 0$
 $x + y = 4$ $xy = 15$
 (c) (i) $x + y = 5$ (ii) $x - y = 4$
 $y = x^2 - 2x + 3$ $y = x^2 + x - 5$

3. Show that the line with equation $x + y = 8$ is tangent to the circle with equation $x^2 - 6x + y^2 - 2y + 2 = 0$. [5 marks]



4. Find the exact values of m for which the line $y = mx + 3$ is tangent to the curve with equation $y = 3x^2 - x + 5$. [5 marks]

5. Let C be the ellipse with equation $4x^2 + 9y^2 = 36$. Find the exact values of k for which the line $2x + 3y = k$ is tangent to C . [6 marks]

6. Show that the line $y = kx + 5$ intersects the parabola $y = x^2 + 2$ for all values of k .
[6 marks]

7. Solve the simultaneous equations.
 $2^x + 2^y = 10$
 $x + y = 4$
 [6 marks]

8. Solve the simultaneous equations.
 $\log_3 x + \log_3 y = 3$
 $y = x^5$
 [6 marks]

4F Systems of linear equations

You already know several methods for solving two linear simultaneous equations with two unknowns. In this section we will extend the method of elimination to deal with three equations and three unknowns. These types of equations will be used later in the course to find the intersection between planes.

Consider the system of equations

$$\begin{cases} 2x + y + z = 7 & (1) \\ x + 3y + 2z = 13 & (2) \\ 4x - 2y + 3z = 9 & (3) \end{cases}$$

Following the strategy we used with two simultaneous equations, we can try to eliminate x . We can choose any two equations; usually we use the second and third. For example, we can multiply the second equation by 2 and subtract it from the first equation:

$$\begin{aligned} 2x + y + z &= 7 & (1) \\ 2 \times (2): 2x + 6y + 4z &= 26 \\ \Rightarrow -5y - 3z &= -19 & (4) \end{aligned}$$

To eliminate x from the third equation, we can multiply (1) by 2 and subtract (3):

$$\begin{aligned} 2 \times (1): 4x + 2y + 2z &= 14 \\ (3): 4x - 2y + 3z &= 9 \\ \Rightarrow 4y - z &= 5 & (5) \end{aligned}$$

Section Q of Prior learning deals with simultaneous equations.

We will study equations of lines and planes in chapter 14.

We can replace equation (2) by equation (4) and equation (3) by equation (5), so now we have a new system of three equations (1), (4) and (5):

$$\begin{cases} 2x + y + z = 7 & (1) \\ -5y - 3z = -19 & (4) \\ 4y - z = 5 & (5) \end{cases}$$

Equations (4) and (5) contain only two unknowns, so we know how to proceed. For example, we can eliminate y if we replace equation (5) by $4 \times (4) + 5 \times (5)$:

$$\begin{aligned} 4 \times (4): -20y - 12z &= -76 \\ 5 \times (5): 20y - 5z &= 25 \\ \Rightarrow -17z &= -51 \quad (6) \end{aligned}$$

The last equation enables us to find z . It is useful to keep track of the equations we will be using to find x and y , so write the system of three equations we now have:

$$\begin{cases} 2x + y + z = 7 & (1) \\ -5y - 3z = -19 & (4) \\ -17z = -51 & (6) \end{cases}$$

Writing the equations like this makes the rest of the strategy clear: We will use equation (6) to find z , then equation (4) to find y and finally equation (1) to find x .

$$(6) \Rightarrow z = 3$$

$$(4) \Rightarrow -5y = -19 + 3 \times 3 \Rightarrow y = \frac{-10}{-5} = 2$$

$$(1) \Rightarrow 2x = 7 - 2 - 3 \Rightarrow x = \frac{2}{2} = 1$$

Thus the solution of our system is $x = 1$, $y = 2$, $z = 3$.

The method carried out above is known as **Gaussian elimination**.

KEY POINT 4.2

To solve a system of linear equations using Gaussian elimination:

First transform the system into a triangular form by eliminating x from the second and third equations and y from the third equation.

Then use back substitution to find z , y and x .

Your calculator can carry out the first part of the procedure. In some cases it can also complete the back substitution to find the solution of the system.

You may know that simultaneous equations do not always have

a solution. For example, the system of equations
$$\begin{cases} x + y = 3 \\ x + y = 2 \end{cases}$$

clearly has no solutions. This can also happen with systems of three equations, and it becomes apparent when the system has been transformed into the triangular form.

EXAM HINT

See calculator sheet 16 for details of Gaussian elimination



Worked example 4.12

Show that the system
$$\begin{cases} x + 2y + 3z = 10 \\ 2x + 3y + 2z = 4 \\ 4x + 7y + 8z = 7 \end{cases}$$
 has no solutions.

We will be able to see whether there are solutions by looking at the triangular form. The calculator can produce this

The last equation is impossible

Using GDC, the system is transformed to

$$\begin{cases} x + \frac{7}{4}y + 2z = \frac{7}{4} \\ y + 4z = -1 \\ 0 = 1 \end{cases}$$

$0 = 1$ is impossible, so there are no solutions.

Systems which have solutions are called **consistent**; otherwise they are **inconsistent**. If a system is consistent, it may have more than one solution. This will always happen when there are more unknowns than equations. For example, the equation $x + y = 5$ has infinitely many solutions: We can pick x to be any number and then $y = 5 - x$. The next example shows what happens with three unknowns.

Worked example 4.13

Find all solutions of the system
$$\begin{cases} 3x - y + 2z = 5 \\ 2x + y + z = 1 \end{cases}$$

Start by eliminating x .

$$\begin{aligned} 2 \times (1): 6x - 2y + 4z &= 10 \\ 3 \times (2): 6x + 3y + 3z &= 3 \\ \Rightarrow -5y + z &= 7 \quad (3) \end{aligned}$$

We cannot do any more elimination, so start on the back substitution. But we can't find y or z from (3), we can only express z in terms of y .

$$\begin{aligned} \Rightarrow z &= 7 + 5y \\ (1) \Rightarrow 3x &= 5 + y + 2(7 + 5y) \\ &= 19 + 11y \\ \therefore x &= \frac{19 + 11y}{3} \end{aligned}$$

We have expressed x and z in terms of y . But y can be any number.

We can write the solution as:
 $x = \frac{19 + 11t}{3}, y = t, z = 7 + 5t$ where $t \in \mathbb{R}$

You will see in chapter 14 that this general solution represents an equation of a straight line, and the system describes two planes intersecting along this line.

In the last example the system has infinitely many solutions. What we have found is called a **general solution**; it is a way of representing all possible solutions.

KEY POINT 4.3

When a system has infinitely many solutions, the **general solution** involves writing x , y and z in terms of a real parameter t .

EXAM HINT

Some calculators can find the general solution so you can use this to check your answer. With others you have to perform back substitution yourself.

The situation with infinitely many solutions can also occur with three equations and three unknowns. The next example also shows you how to set out your working when performing Gaussian elimination.

Worked example 4.14

- (a) Find the value of k for which the system of equations
$$\begin{cases} 2x + y + 2z = 4 \\ 6x - y - 6z = 8 \\ 4x + 6y + 16z = k \end{cases}$$
 is consistent.
- (b) For the value of k found above, find the general solution of the system.

Do Gaussian elimination. Start by labelling all equations

$$\begin{aligned} \text{(a)} \quad & \begin{cases} 2x + y + 2z = 4 & (1) \\ 6x - y - 6z = 8 & (2) \\ 4x + 6y + 16z = k & (3) \end{cases} \end{aligned}$$

Eliminate x from second and third equations

$$\begin{aligned} & \begin{cases} 2x + y + 2z = 4 & (1) \\ 3 \times (1) - (2) & \begin{cases} 4y + 12z = 4 & (4) \\ -4y - 12z = 8 - k & (5) \end{cases} \end{cases} \end{aligned}$$

Eliminate y from the third equation

$$\begin{aligned} & \begin{cases} 2x + y + 2z = 4 & (1) \\ 4y + 12z = 4 & (4) \\ (4) + (5) & \begin{cases} 0 = 12 - k & (6) \end{cases} \end{cases} \end{aligned}$$

For the system to be consistent, the last equation must say $0 = 0$

The system is consistent when $k = 12$.

Start back substitution from the middle equation. We can express y in terms of z

$$\begin{aligned} \text{(b)} \quad & \text{Let } z = t \\ & (4) \Rightarrow y = \frac{4 - 12t}{4} \\ & \quad = 1 - 3t \\ & (1) \Rightarrow 2x = 4 - (1 - 3t) - 2t \\ & \quad = 3 + t \\ & \Rightarrow x = \frac{3 + t}{2} \end{aligned}$$

We now have a general solution

$$\begin{aligned} & \text{The general solution is} \\ & x = \frac{3 + t}{2} \\ & y = 1 - 3t \\ & z = t, t \in \mathbb{R} \end{aligned}$$

The method of Gaussian elimination can be extended to systems with more equations and more unknowns.



EXAM HINT

Remember that when there is a unique solution, you can also find it using the simultaneous equation solver. On some calculators the equation solver can also find the general solution. See calculator sheet xx



When performing Gaussian elimination without a calculator it is useful to look for shortcuts. In the above example, we could have divided equation (4) by 4 before continuing. You can also change the order of the equations, for example so that the first one has the smallest coefficient of x . If the first equation does not contain x at all, you need to arrange them so that the first equation is one that does. Note that your calculator will usually divide equations by a number so that the coefficients on the diagonal are all 1, but there is no need for you to do this.

We can summarise all the possibilities for a system of three linear equations with three unknowns:

KEY POINT 4.4

The number of solutions of a system of three linear equations in three unknowns can be seen from the triangular form.

If the last equation has the form $kz = c$ with $k \neq 0$, there is a unique solution for x , y and z .

If the last equation has the form $0 = c$ with $c \neq 0$, there are no solutions (the system is inconsistent).

If the last equation is $0 = 0$, there are infinitely many solutions. The general solution can be found by setting $z = t$ and expressing x and y in terms of t .

The next example illustrates all the possibilities.

Worked example 4.15

Consider the system of equations:

$$\begin{cases} 3x + 2y - 2z = 1 \\ 2x - 4y + z = 4 \\ x + y + kz = a \end{cases}$$

- Find the set of values of k for which the system has a unique solution.
- For the value of k for which the system does not have a unique solution, find the value of a for which the system has infinitely many solutions.
- State the set of values of k and a for which the system has no solutions.



continued...

Start by performing Gaussian elimination

$$\begin{aligned} (a) \quad & \begin{cases} 3x + 2y - 2z = 1 & (1) \\ 2x - 4y + z = 4 & (2) \\ x + y + kz = a & (3) \end{cases} \end{aligned}$$

$$\begin{aligned} 2 \times (1) - 3 \times (2) \quad & \begin{cases} 3x + 2y - 2z = 1 & (1) \\ 16y - 7z = -10 & (4) \end{cases} \\ (1) - 3 \times (3) \quad & \begin{cases} -y - (2 + 3k)z = 1 - 3a & (5) \end{cases} \end{aligned}$$

$$\begin{aligned} (4) + 16 \times (5) \quad & \begin{cases} 3x + 2y - 2z = 1 & (1) \\ 16y - 7z = -10 & (4) \\ -(39 + 48k)z = 6 - 48a & (6) \end{cases} \end{aligned}$$

For a unique solution, the coefficient of z in the last equation must not be zero

Unique solution when $39 + 48k \neq 0$

$$\Leftrightarrow k \neq -\frac{39}{48}$$

When the LHS side of the last equation is zero, the RHS must be zero as well

$$\begin{aligned} (b) \quad & \text{When } k = -\frac{39}{48}, \text{ the system is} \\ & \text{consistent when } 6 - 48a = 0 \\ & \Leftrightarrow a = \frac{1}{8} \end{aligned}$$

No solutions when in the last equation LHS is zero but RHS is not

$$\begin{aligned} (c) \quad & \text{There are no solutions when} \\ & k = -\frac{39}{48} \text{ and } a \neq \frac{1}{8}. \end{aligned}$$

In chapter 14 you will see that the different cases for the number of solutions of a system of three equations correspond to different possibilities for the relative position of three planes in space.

Exercise 4F

1. Using Gaussian elimination, or otherwise, solve the following systems of equations:

$$(a) \quad (i) \quad \begin{cases} 2x - 3y + z = 10 \\ 5x + 2y + 2z = 15 \\ x + 4y - 2z = -3 \end{cases}$$

$$\begin{aligned}
 & \text{(ii)} \quad \begin{cases} x - y + 3z = -1 \\ 2x + y + z = -1 \\ x - 2y + 3z = -3 \end{cases} \\
 & \text{(b) (i)} \quad \begin{cases} 2x + y + z = 3 \\ x - y + 3z = 6 \\ 4x + 3y + z = 3 \end{cases} \\
 & \text{(ii)} \quad \begin{cases} x - 3y + z = -5 \\ 3x - y - z = -7 \\ x + 3y + 5z = 1 \end{cases} \\
 & \text{(c) (i)} \quad \begin{cases} 2y + 3z = 4 \\ x + 3y + 2z = 4 \\ -2x + y + 3z = -1 \end{cases} \\
 & \text{(ii)} \quad \begin{cases} y - 3z = -9 \\ 2x + y + z = 1 \\ x - 3y + z = 2 \end{cases}
 \end{aligned}$$



2. Use your calculator to transform the following systems into a triangular form, and hence say whether they have a unique solution, no solutions, or infinitely many solutions.

$$\begin{aligned}
 & \text{(a)} \quad \begin{cases} 3x - y + 4z = 7 \\ x - 2y + z = 3 \\ x - y + 4z = -5 \end{cases} \\
 & \text{(b)} \quad \begin{cases} 2x - y - z = 5 \\ 3x + 2y + 3z = 1 \\ 5x + y + 2z = 4 \end{cases} \\
 & \text{(c)} \quad \begin{cases} -2x + y + 3z = 5 \\ 2x + y + z = 2 \\ 2x + 3y + 5z = 9 \end{cases} \\
 & \text{(d)} \quad \begin{cases} x - 2y + z = 4 \\ -2x + 4y - 2z = 1 \\ 3x - 6y + 3z = 12 \end{cases} \\
 & \text{(e)} \quad \begin{cases} x - 2y + z = 0 \\ 3x + 2y + 2z = 0 \\ x - 5y - z = 0 \end{cases}
 \end{aligned}$$

3. Find the general solution of the following systems of equations:

$$(a) (i) \begin{cases} 2x + y + z = 5 \\ x - y + 2z = 1 \\ x + 2y - z = 4 \end{cases}$$

$$(ii) \begin{cases} 2x - y - 3z = 3 \\ x + y - 3z = 0 \\ x + 2y - 4z = -1 \end{cases}$$

$$(b) \begin{cases} x - 2y + z = 1 \\ 3x - 6y + z = 3 \\ -x + 2y + 2z = -1 \end{cases}$$

$$(c) \begin{cases} x - 2y + z = 5 \\ -3x + 6y - 3z = -15 \\ 2x - 4y + 2z = 10 \end{cases}$$

4. Find the value of c for which the following system of equations is consistent:

$$\begin{cases} 2x + y - 2z = 0 \\ x - 2y - z = 2 \\ 3x + 4y - 3z = c \end{cases}$$

[5 marks]

5. For which value of k does the system of equations

$$\begin{cases} x - 2y + 2z = 0 \\ 2x + ky - z = 3 \\ x - y + 3z = -5 \end{cases}$$

not have a unique solution?

[6 marks]

6. Find the solution of the following system of equations:

$$\begin{cases} 2x + y - z = 2 \\ x - 2y + 2z = 1 \\ 2x + y - 4z = a \end{cases}$$

[7 marks]



7. (a) Find the values of a for which the following system of equations is consistent:

$$\begin{cases} x - 2y + z = 1 \\ 2x + y - z = a \\ 4x + 7y - 5z = a^2 \end{cases}$$

- (b) For the larger of the two values found above, find the general solution of the system. [10 marks]

8. (a) Find the value of k for which the system of equations

$$\begin{cases} x - 2y + z = 7 \\ 2x + y - 3z = b \\ x + y + kz = 4 \end{cases}$$

does not have a unique solution.

- (b) For this value of k , find the value of b for which the system is consistent.
(c) In the case when the system has infinitely many solutions, find the general solution. [11 marks]

4G Solving inequalities

You already know that:

- you can add or subtract the same number from both sides of an **inequality** and multiply or divide both sides by a positive number
- if you multiply or divide both sides by a negative number you must reverse the sign.

How about using the rule 'do the same thing to both sides' with other functions we will meet in this course?

You know from Prior learning that the solution to $x^2 > 4$ is *not* $x > 2$ or even $x > \pm 2$; it is in fact $x < -2$ or $x > 2$. This is easiest to see when you sketch the graph and look for regions which satisfy the inequality.

Inequalities with other functions can be even more complicated.

For example, trying some numbers reveals that the inequality

$$\left(\frac{1}{2}\right)^x < 4 \text{ is satisfied for } x > -2, \text{ while the inequality } x^3 - 4x > 2$$

is satisfied for (among other numbers) $x = -1, 3, 4$ but not for $x = -2, 0, 1, 2$.

Given the difficulties with rearranging inequalities, it is often best to apply only simple operations and otherwise solve them graphically. Inequalities can appear on the non-calculator paper; there are many graphs that you must know how to sketch without a calculator.

EXAM HINT

Whenever solving inequalities always sketch the graph!

Worked example 4.16

Solve the following inequality $(x-2)(x^2-5x+5) > x-2$.

We are going to try to get everything onto one side and then factorise it so that we can draw the graph.

Notice that there is a factor of $x-2$ on both sides. We can use this to make the algebra neater

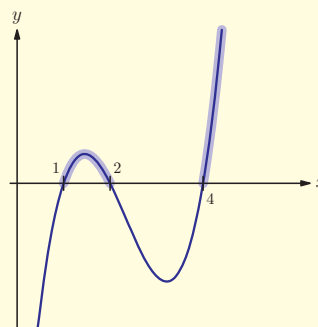
Sketch the cubic graph, highlighting where it is above the axis

$$\Leftrightarrow (x-2)(x^2-5x+5) - 1(x-2) > 0$$

$$\Leftrightarrow (x-2)(x^2-5x+5-1) > 0$$

$$\Leftrightarrow (x-2)(x^2-5x+4) > 0$$

$$(x-2)(x-1)(x-4) > 0$$



$$\therefore 1 < x < 2 \text{ or } x > 4$$

EXAM HINT

If you are describing distinct regions you must combine the inequalities using 'or'. You can only write a single inequality, such as $-6 \leq x \leq 1$, if it describes one region.

Note that the answer from the last example can also be written using interval notation as $x \in]1, 2[\cup]4, \infty[$. You do not have to use this notation, but you must be able to understand it.



See Prior learning I for an explanation of interval notation.

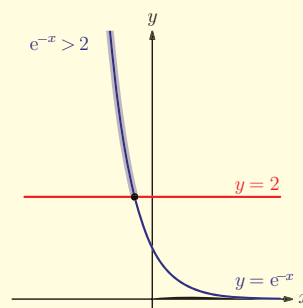
Worked example 4.17

 Solve the inequality $e^{-x} > 2$.

Sketch the graph of $y = e^{-x}$ and look to see where it is above the line $y = 2$

Find the intersection point

The highlighted part is to the left of the intersection point



$$\begin{aligned} \text{Intersection: } e^{-x} &= 2 \\ \Leftrightarrow -x &= \ln 2 \\ \Leftrightarrow x &= -\ln 2 \end{aligned}$$

$$\text{So } x < -\ln 2$$

EXAM HINT

If you are going to use the calculator there is no need to rearrange the inequality in any way – just plot the two sides.

Exercise 4G

1. Solve the following inequalities:

- | | |
|--|--|
| (a) (i) $2.4^x < 14.8$ | (ii) $3.1^x > 6$ |
| (b) (i) $0.8^x \geq 0.4$ | (ii) $0.4^x \leq 0.001$ |
| (c) (i) $3 \times \left(\frac{3}{4}\right)^{2x} > 1$ | (ii) $5 \times \left(\frac{4}{3}\right)^{3x+4} < 30$ |
| (d) (i) $3x^5 \geq 14$ | (ii) $4x^3 + 1 < 6$ |
| (e) (i) $\ln(x+4) > 0.8$ | (ii) $\ln(x-3) \leq 0.4$ |
| (f) (i) $x^3 - 4x > 0$ | (ii) $x^3 - 6x < 0$ |

2. By sketching both sides on the same diagram, solve the following inequalities:

(a) $x > 5 \ln x$ (b) $\frac{x}{\ln x} > 5$

Why are the answers not the same?

[5 marks]

4H Working with identities

An identity is an equation which is true for all values of x , for example $x \times x \equiv x^2$.

The identity sign $\equiv$ is used to emphasise the identity, although often an equals sign is used instead, with the words 'for all x '.

You have already seen in Section 3A that one use of identities is the ability to compare coefficients. Another important use of identities is to manipulate expressions.

Operations such as opening out brackets are actually applications of identities. For example when manipulating the expression $(x - 2)(x + 1)$ to give $x^2 - x - 2$ we are actually using the identity $(x + a)(x + b) \equiv x^2 + (a + b)x + ab$ for all a , b and x .

To prove an identity we must start from one side and state which identities or rules are being used as we transform it line by line to reach the other side of the identity.

EXAM HINT

You may be unsure about which rules and identities you are allowed to use. Anything listed as a Key point in this book is acceptable, as are basic algebraic manipulations such as multiplying out brackets and simplifying fractions.

Worked example 4.18

Prove that $e^{(2 \ln x + \ln 3)} = 3x^2$.

Try to simplify the exponent using laws of logarithms

First, take the multiple inside the log: $p \ln a = \ln(a^p)$

Then $\ln a + \ln b = \ln(ab)$

Finally, apply the cancellation principle

$$\text{LHS} = e^{(2 \ln x + \ln 3)}$$

$$= e^{\ln x^2 + \ln 3}$$

$$= e^{\ln(3x^2)}$$

$$= 3x^2$$

$$= \text{RHS}$$

There are other ways to prove identities.

- We are allowed to work from both ends and meet in the middle.

Proving identities is very important in trigonometry – see chapter 10.

- We can subtract one side from the other and show that this is always 0.
- We can divide one side by the other and show that this is always 1.

It can be very tempting to treat the identity to be proven like an equation and do things to both sides until we get something which we know is correct. However, this is logically flawed and you should avoid it.

Worked example 4.19

Prove that for all x ,

$$\frac{2}{x} - \frac{1+x}{x^2-x} = \frac{3}{x} - \frac{2}{x-1}$$

We must start from one side and try to get to the other. The left-hand side looks a little more complicated, so start there. See if we can simplify

$$\begin{aligned} \text{LHS} &= \frac{2}{x} - \frac{1+x}{x^2-x} \\ &= \frac{2}{x} - \frac{1+x}{x(x-1)} \\ &= \frac{2(x-1) - (1+x)}{x(x-1)} \\ &= \frac{x-3}{x(x-1)} \end{aligned}$$

There is no obvious way to continue, so now start from the right-hand side and see if we can get to the same place

$$\begin{aligned} \text{RHS} &= \frac{3}{x} - \frac{2}{x-1} \\ &= \frac{3(x-1) - 2x}{x(x-1)} \\ &= \frac{x-3}{x(x-1)} \end{aligned}$$

We have shown that the two sides are equal

LHS=RHS, so the identity is proved.

Exercise 4H

1. Prove the following identities.

(a) (i) $(x-y)^2 + 4xy = (x+y)^2$

(ii) $x^3 + y^3 = (x+y)(x^2 - xy + y^2)$

(b) (i) $\log(xyz) = \log x + \log y + \log z$ (for $x, y, z > 0$)

$$(ii) \log_a b = \frac{1}{\log_b a} \text{ (for } a, b > 0, a, b \neq 1)$$

$$(c) (i) \frac{a^2 - b^2}{a - b} = a + b \quad (ii) \frac{x - y}{y - x} = -1$$

$$(d) (i) \sqrt[a]{x^b} = (\sqrt[a]{x})^b \quad (ii) 2^a + 2^a = 2^{a+1}$$

2. If the cubic expression $x^3 + px + q$ can be written in the form $(x - a)^2(x - b)$, find an equation linking p and q . [6 marks]

Summary

- Important methods for solving equations include factorising and substitution.
- When sketching graphs on your calculator, you may need to use your knowledge of the shapes of common graphs to make sure you do not miss any important features, such as zeros, vertical or horizontal asymptotes.
- You can use a graphical calculator to solve equations involving unfamiliar functions, by finding intersections of two curves.
- Systems of three equations in three unknowns can be solved using **Gaussian elimination**. The system may have a unique solution, infinitely many solutions, or no solutions.
- When a system of equations has infinitely many solutions, a **general solution** has x, y, z expressed in terms of a real parameter t .
- Sketching graphs is the best way to solve complicated **inequalities**.
- **Identities** are true for all values of x . We can prove identities by transforming one side into the other, using known identities.

Introductory problem revisited

What information can be extracted from the equation $ax + b = 3x + 2$?
 What information can be extracted from the identity $ax + b = 3x + 2$ for all x ?

$ax + b = 3x + 2$ is an equation, so it will only be true for some values of x .

Rearranging gives the answer $x = \frac{2 - b}{a - 3}$.

$ax + b \equiv 3x + 2$ is an identity, so it is true for all x . Comparing coefficients gives $a = 3$ and $b = 2$.

The equation allows us to examine x in terms of a and b .

The identity allows us to evaluate a and b .

Mixed examination practice 4

Short questions



1. (a) Sketch the graphs $y = 2^x$ and $y = 1 - x^2$ on the same axes.

(b) Hence write down the number of solutions of the equation $2^x = 1 - x^2$.

[6 marks]



2. Find the largest possible value of $y = x^2 e^{-x}$ for $x \in [0, 5]$.

[4 marks]



3. Solve the inequality $x^3 - 4x > e^{-x}$.

[5 marks]



4. Solve the following system of equations:

$$\begin{cases} 3x - y + 2z = 2 \\ x - 2y + z = 3 \\ x - y + 3z = -5 \end{cases}$$

[6 marks]

5. Find the exact solution of the equation $e^x \ln x = 3e^x$.

[5 marks]



6. (a) By using an appropriate substitution find the exact solutions to the equation $x^4 + 36 = 13x^2$.

(b) Hence solve the inequality $x^4 + 36 \leq 13x^2$.

[6 marks]



7. (a) Sketch the graph of $\frac{1}{e^x - 2}$.

(b) State the exact equation of the vertical asymptote.

[6 marks]

8. (a) Find the value of p for which the following system of equations is consistent:

$$\begin{cases} 2x + 2y - z = 1 \\ x + y - z = -4 \\ 4x + 4y - 3z = p \end{cases}$$

(b) For this value of p , find the general solution of the system.

[9 marks]



9. Find the values of k for which the line $y = 2x - k$ is tangent to the circle with equation $x^2 + y^2 = 5$.



10. Solve the inequality $\frac{x}{\ln x} < 4$.

[5 marks]



11. Solve the equation $x \ln x + 4 \ln x = 0$.

[5 marks]

Long questions

1. (a) (i) Find the value of k for which the equation $x^3 - 9x^2 + kx - 28 = 0$ can be written in the form $(x - a)^3 = b$.
 (ii) For this value of k , find the exact solution of the equation $x^3 - 9x^2 + kx - 28 = 0$.
 (b) (i) Sketch the graph of $\frac{1}{(x-3)^3 - 1}$.
 (ii) Write down the equations of the asymptotes.
 (c) Find the coordinates of the minimum point on the graph of $y = (x-3)^3 \ln x$. [10 marks]

2. (a) If $f(x) = x^4 + x^2 - 6$ solve the equation $f(x) = 0$.
 (b) (i) Show that the equation $x^4 + 7x^2 = 4x^3 + 6x + 4$ can be written in the form $f(x+k) = 0$ where k is a constant to be determined.
 (ii) Hence find the exact solutions to the equation $x^4 + 7x^2 = 4x^3 + 6x + 4$.
 (c) Solve the inequality $x^4 + 7x^2 > 4x^3 + 6x + 4$. [12 marks]

3. (a) Find the value of k for which the system of equations

$$\begin{cases} x + y + z = 3 \\ x + ky + z = 4 \\ x - y + z = b \end{cases}$$
 does not have a unique solution.
 (b) For this value of k , find the value of b for which the system is consistent.
 (c) In the case when the system has infinitely many solutions, find the general solution. [11 marks]